

Immunodeficiency 122 (IMD122) — Comprehensive Disease Characterization

Disease: Immunodeficiency 122 (IMD122) **Causal gene:** *POLD3* (DNA polymerase delta 3, accessory subunit) **Identifiers:** OMIM:620869 · MONDO:0971151 · DOID:0061088 · MedGen C5935632 (CUI) · UMLS C5935632 **Evidence base:** Ultra-rare Mendelian disorder described in only two unrelated patients (2023). All clinical content is derived from **individual patient reports** (case-level), not aggregated registries, plus in vitro functional and mouse model data.

Summary (Answer to the Research Question)

Immunodeficiency 122 is an **ultra-rare autosomal-recessive syndromic combined immunodeficiency** caused by **biallelic hypomorphic missense variants in *POLD3***, which encodes the p66/p68 accessory, complex-stabilizing subunit of **DNA polymerase δ** (and a shared subunit of translesion polymerase ζ). Reduced polymerase δ function produces **replicative stress, defective S-phase progression, and accumulation of DNA double-strand breaks**, which impair proliferation and V(D)J/TCR recombination in lymphoid progenitors—producing **severe combined immunodeficiency (SCID), frequently presenting as Omenn syndrome**—and injure other highly proliferative lineages, causing **neurodevelopmental delay, sensorineural hearing loss, and ectodermal anomalies**. Onset is neonatal/congenital with a severe course; hematopoietic stem cell transplantation can reconstitute immunity but does not correct the cell-intrinsic replication defect in non-hematopoietic tissues.

1. Disease Information

- **Overview:** IMD122 is a monogenic inborn error of immunity in the category of **combined immunodeficiencies with syndromic/associated features** (per IUIS classification). It combines a profound T-cell ($\pm$ B/NK) defect with neurodevelopmental, auditory, and ectodermal manifestations due to a defect in genome replication/maintenance.
- **Key identifiers:**

- OMIM phenotype: **620869** (Immunodeficiency 122)
- MONDO: **MONDO:0971151**
- DOID: **0061088**; MedGen/UMLS: **C5935632**
- Orphanet: no dedicated code identified (newly described; would fall under "Combined immunodeficiency")
- ICD-11: maps to **4A00** (Primary immunodeficiencies) / ICD-10 **D81.x** (combined immunodeficiencies)
- MeSH: no specific descriptor; closest = *Severe Combined Immunodeficiency* (D016511); phenotype also = *Omenn syndrome*
- **Synonyms / alternative names:** IMD122; **POLD3 deficiency**; DNA polymerase delta 3 deficiency; syndromic SCID due to POLD3; autosomal-recessive polymerase δ (PolD) deficiency (POLD3 type); POLD3-related Omenn syndrome.
- **Evidence source:** Individual-patient (case report) level — two probands total

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2. Etiology

- **Causal factor (genetic):** Biallelic (homozygous) **pathogenic missense variants in POLD3** (AR). Both reported patients were offspring of **consanguineous unions**.
- "*Here we report a homozygous POLD3 variant (NM_006591.3; p.Ile10Thr) in a Lebanese patient, the product of a consanguineous family, presenting with a syndromic severe combined immunodeficiency (SCID)*"

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- **Genetic risk factors:** The causal variants themselves (p.Ile10Thr; p.K373T). No susceptibility loci or modifier genes identified (too few cases). **Consanguinity** is the principal risk context. POLD3 is strongly **LoF-intolerant** (gnomAD pLI 0.994, LOEUF 0.48), so only **hypomorphic missense** alleles—retaining partial function—cause viable disease; complete LoF is presumed lethal (mouse null embryonic-lethal,

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- **Environmental risk factors:** None established; disease is fully genetically determined. Infectious exposures are **consequences** (opportunistic infection due to immunodeficiency), not causes.
- **Protective factors:** None reported (genetic or environmental). Not applicable at the population level given rarity.
- **Gene–environment interactions:** Not characterized. Mechanistically, exogenous genotoxic/replication stressors (UV, chemotherapeutics, oxidative stress) would be predicted to aggravate the underlying replication defect (inferred, not demonstrated).

3. Phenotypes

Onset **neonatal/congenital**, severity **severe**, course **progressive**. Frequencies are qualitative (n=2). HPO terms derived from OMIM:620869 annotation.

Immune / laboratory abnormalities: - Severe combined immunodeficiency — **HP:0000958** - Decreased total T cells — **HP:0500093**; decreased naive CD4+ (**HP:0001744**) and naive CD8+ (**HP:0005359**) proportions; oligoclonal T-cell expansion — **HP:0010975** - Reduced NK cell count — **HP:0000407**; abnormal B-cell count — **HP:0410243**; decreased unswitched memory B cells — **HP:0040218** - Decreased IgG — **HP:0001019**; abnormal IgM — **HP:0004315**; **increased IgE** — **HP:0002716**; eosinophilia — **HP:0001999** - Thymic aplasia/hypoplasia — **HP:0004430**

Omenn-syndrome features (clinical signs/physical): - Erythroderma — **HP:0410377**; alopecia — **HP:0008404**; lymphadenopathy — **HP:0002110**; hepatomegaly — **HP:0004429**; splenomegaly — **HP:0002783**

Infections (symptoms): recurrent viral — **HP:0002788**; bacterial — **HP:0001270**; upper (**HP:0001263**) and lower (**HP:0032126**) respiratory infections; bronchiectasis — **HP:0002007**

Neurodevelopmental (behavioral/developmental): global developmental delay — **HP:0001596**; motor delay — **HP:0002718**; (progressive neurological regression in the Omenn patient)

Sensory: sensorineural hearing impairment — **HP:0001880**

Craniofacial / ectodermal (physical): frontal bossing — **HP:0003212**; abnormal facial shape — **HP:0410378**; enamel hypoplasia — **HP:0011968**; nail dystrophy — **HP:0031430**; dry skin — **HP:0005403**

Feeding/allergy: feeding difficulties — **HP:0002240**; food allergy — **HP:0006297**

Quality-of-life impact: profound—life-threatening infections, failure to thrive, developmental disability, deafness; without HSCT, SCID is uniformly fatal in infancy. Formal QoL instruments (EQ-5D/SF-36) not applicable/available.

4. Genetic / Molecular Information

- **Causal gene: *POLD3*** — HGNC:20932; NCBI Gene **10714**; Ensembl **ENSG00000077514**; cytoband **11q13.4** (GRCh38 chr11:74,493,851–74,669,117, + strand); transcript **NM_006591.3**; protein **UniProt Q15054** (466 aa, "DNA polymerase delta subunit 3", p66/p68).

- **Pathogenic variants (both germline, homozygous, missense):** | Variant (cDNA/protein) | Patient | Classification | Consequence | |---|---|---|---| | NM_006591.3:c.29T>C, **p.Ile10Thr** | Lebanese, syndromic SCID

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| Pathogenic (AR) | Abolishes **POLD3 and POLD1/POLD2** protein expression → complex destabilization | | **c.1118A>C, p.Lys373Thr (p.K373T)** | Omenn syndrome

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| Pathogenic (AR) | Normal PolD subunit levels but defective function → S-phase defect, ↑ dsDNA breaks (rescued by WT **POLD3**) |

- **Variant type/class:** missense (both). No nonsense/frameshift/splice/structural variants reported (consistent with LoF intolerance).
- **Allele frequency:** both variants are **private/ultra-rare**, essentially absent from gnomAD; gene-level LoF constraint pLI **0.994**, LOEUF **0.48**, LoF o/e **0.33** (missense unconstrained, mis_z 0.76).
- **Somatic vs germline:** germline.
- **Functional consequence: loss/reduction of function (hypomorphic).** p.Ile10Thr → destabilization of the entire Pol δ heterocomplex; p.K373T → intrinsic functional impairment with preserved assembly.
- **Modifier genes:** none identified.
- **Epigenetic information:** none reported for this disorder.
- **Chromosomal abnormalities:** none (point mutations only). *Downstream*, **POLD3** deficiency causes acquired genomic instability (chromosome breaks, micronuclei, aneuploidy —

shown in mouse/cell models,

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5. Environmental Information

- **Environmental factors:** none causative. IMD122 is a monogenic disorder.
- **Lifestyle factors:** not applicable (congenital, infantile-lethal).
- **Infectious agents:** not causal but central to morbidity—patients suffer **opportunistic and recurrent viral, bacterial, and respiratory infections** secondary to the immunodeficiency (HP:0002788/HP:0001270/HP:0032126). Typical SCID pathogens (inferred from SCID literature): *Pneumocystis jirovecii*, CMV, adenovirus, RSV, candida; live-vaccine organisms (BCG, rotavirus) are contraindicated.

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation):

1. **Biallelic hypomorphic missense variant in *POLD3*** (p.Ile10Thr or p.K373T) → **reduced functional DNA polymerase δ** , either by destabilizing/degrading the POLD1–POLD2–POLD3(–POLD4) complex (p.Ile10Thr abolishes POLD1/POLD2/POLD3 expression) or by impairing complex function with preserved subunit levels (p.K373T). (*Demonstrated in vitro.*)
2. Reduced Pol δ activity → **impaired high-fidelity DNA replication** (leading- and lagging-strand synthesis, loss of PCNA-stimulated processivity) and reduced DNA repair capacity. (*Demonstrated — UniProt-annotated functions; Conde POLD1/POLD2 data*

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3. → **Replicative stress: defective S-phase entry/progression and accumulation of DNA double-strand breaks (γ H2AX foci)**; the defect is **rescued by wild-type POLD3**, establishing causality. (*Demonstrated —*

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analogous mouse data

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extended S-phase, telomere loss, chromosome breaks, micronuclei, aneuploidy, DDR via ATR/ATM/53BP1/RIF1.) 4a. **Immune branch:** replication stress plus a **defect in the early stages of TCR recombination** → failure of thymocyte proliferation/V(D)J recombination → **profound naive T-cell lymphopenia with restricted/oligoclonal TCR repertoire** → **SCID**; residual autoreactive oligoclonal T-cell expansion with Th2 skewing → **Omenn syndrome** (erythroderma, eosinophilia, high IgE, hepatosplenomegaly, lymphadenopathy).

(Demonstrated —

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4b. **Neurodevelopmental branch:** replication stress in neural progenitors → impaired neurogenesis → **global developmental delay / progressive neurological regression.**

(Inferred by analogy to POLD1/POLD2;

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4c. **Auditory/ectodermal branch:** replication stress in cochlear and ectodermal (skin, hair, nail, dental enamel) progenitors → **sensorineural hearing loss and ectodermal dysplasia-like features.** *(Inferred.)*

4. → **Clinical syndrome:** neonatal-onset syndromic SCID/Omenn with neurodevelopmental delay, deafness, and ectodermal/craniofacial anomalies; death in infancy/early childhood without—and sometimes despite—HSCT.

Supporting detail: - **Molecular pathways/processes:** DNA replication (GO:0006260), DNA repair / double-strand-break repair (GO:0006302), replication-stress / DNA-damage response via ATR–ATM–53BP1–RIF1, cell-cycle S-phase (GO:0000082), V(D)J recombination (GO:0033151), T-cell differentiation (GO:0030217), apoptosis (GO:0006915). Reactome: "DNA strand elongation," "Polymerase switching," "Lagging strand synthesis." KEGG: *DNA replication* (hsa03030), *Base/Nucleotide excision repair*, *Mismatch repair*. - **Protein dysfunction:** loss/reduction of function of the Pol δ accessory subunit; POLD3 also co-forms **DNA polymerase ζ** (translesion synthesis, with REV3L/REV7) and participates in break-induced replication/homologous recombination—so its loss broadly compromises genome maintenance. - **Cellular processes:** cell-cycle dysregulation (extended/arrested S phase), genomic instability, apoptosis of stressed progenitors. - **Immune involvement:** this is a primary **immunodeficiency** (combined T/B/NK), not autoimmunity per se—though Omenn physiology produces immune dysregulation/Th2 inflammation. - **Cell types (CL):** T cells (CL:0000084), naive T cell (CL:0000898), thymocyte (CL:0000893), double-positive thymocyte (CL:0000809), B cell

(CL:0000236), natural killer cell (CL:0000623), hematopoietic stem cell (CL:0000037), neural progenitor/neuron (CL:0000031/CL:0000540), keratinocyte (CL:0000312). - **Molecular profiling:** no transcriptomic/proteomic/metabolomic datasets published for POLD3 patients; functional readouts to date are flow-cytometric immunophenotyping, TCR-repertoire sequencing, cell-cycle/EdU S-phase assays, and γ H2AX DSB-foci quantification.

7. Anatomical Structures Affected

- **Organ level (primary):** **thymus** (aplasia/hypoplasia, UBERON:0002370), **bone marrow / hematopoietic system** (UBERON:0002371), **lymph nodes / spleen** (lymphadenopathy, splenomegaly; UBERON:0000029/UBERON:0002106), **brain / CNS** (UBERON:0000955), **inner ear / cochlea** (UBERON:0001846/UBERON:0001844).
 - **Secondary involvement:** **lungs/airways** (recurrent infection, bronchiectasis; UBERON:0002048), **liver** (hepatomegaly; UBERON:0002107), **skin** (erythroderma, dryness; UBERON:0002097), **hair follicles** (alopecia), **nails, teeth/enamel**.
 - **Body systems:** immune/lymphatic, nervous, auditory, integumentary, respiratory (secondary), digestive (feeding difficulties).
 - **Tissue/cell level:** lymphoid tissue and lymphocyte progenitors; neural tissue; cochlear/sensory epithelium; ectodermal epithelia (epidermis, hair, nail matrix, ameloblasts).
 - **Subcellular (GO Cellular Component):** **nucleus** (GO:0005634), **delta DNA polymerase complex** (GO:0043625), **zeta DNA polymerase complex** (GO:0016035), replication fork (GO:0005657). POLD3 localizes to nucleus and cytoplasm (UniProt Q15054).
 - **Lateralization:** systemic/bilateral (e.g., bilateral sensorineural hearing loss); not lateralized.
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8. Temporal Development

- **Onset:** congenital/**neonatal**; immunologic and syndromic features manifest in early infancy. Onset pattern **acute to subacute** for infections (SCID presentation), with an **insidious-progressive** neurodevelopmental component.
- **Progression:** rapid for the immune defect; **progressive** neurological regression (documented post-HSCT in the Omenn patient). Disease course is **progressive**, not relapsing/episodic.

- **Duration/stages:** without curative therapy, SCID is fatal in the first 1–2 years; with HSCT, immune reconstitution is possible but **non-immune (neuro/auditory) disease continues to progress** → death reported at **age 4 years** despite HSCT at 6 months

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- **Remission patterns:** no spontaneous remission; partial, compartment-specific improvement with HSCT (immune only).
- **Critical period / window of intervention:** the neonatal–early-infancy window (ideally **pre-infection HSCT**, <3.5 months as in classic SCID) is critical for the immune compartment; however, the replication defect in CNS/cochlea likely has a developmental origin that transplantation cannot reverse.

9. Inheritance and Population

- **Inheritance: Autosomal recessive** (biallelic/homozygous). Both reported families **consanguineous**.
 - **Penetrance:** presumed **complete** for biallelic pathogenic genotypes (n=2); **expressivity** variable between the two reports (classic SCID vs. Omenn presentation).
 - **Anticipation:** not applicable (not a repeat-expansion disorder). **Germline mosaicism:** not reported.
 - **Founder effects / variant geography:** both variants are **private** (one Lebanese, one in a separate pedigree reported from Europe); no founder haplotype established. Consanguinity (e.g., Middle Eastern/North African populations) increases the chance of homozygosity.
 - **Carrier frequency:** not established; both alleles are essentially absent from gnomAD, so heterozygous carriers are vanishingly rare.
 - **Epidemiology: ultra-rare**—only **two patients reported worldwide** (2023). Prevalence/incidence not quantifiable (far below 1/1,000,000). No sex predilection established (both reported probands male, but n too small to infer a sex ratio); AR inheritance predicts equal sex distribution. Age distribution: infants/young children.
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10. Diagnostics

- **Laboratory tests:**

- Lymphocyte subsets by flow cytometry: **low total/naive T cells** (low TRECs), variable B and **low NK cells**; reduced naive CD4+/CD8+.
- Immunoglobulins: **low IgG**, abnormal IgM, **elevated IgE**; **eosinophilia** (CBC with differential).
- TCR-repertoire analysis: **restricted/oligoclonal** repertoire; evidence of defective early TCR recombination.
- **Biomarkers:** no specific circulating biomarker; **cellular replication-stress markers** (γ H2AX DSB foci; defective EdU/S-phase incorporation in patient fibroblasts) are diagnostic functional supports

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- **Imaging / functional:** absent thymic shadow on chest imaging; **audiometry/ABR** for sensorineural hearing loss; brain MRI for neurodevelopmental assessment; chest CT for bronchiectasis.
- **Biopsy/pathology:** Omenn skin shows erythroderma; lymphoid tissue shows oligoclonal T-cell infiltration (consistent with Omenn; general).
- **Genetic testing (diagnostic gold standard):**
- **Whole-exome (WES) or whole-genome (WGS)** sequencing identified both cases; **IEI/SCID/combined-immunodeficiency NGS gene panels** should include **POLD1, POLD2, POLD3**.
- **Single-gene/targeted** confirmation and **cascade/segregation** testing in families; homozygosity mapping is useful in consanguineous pedigrees.
- CMA/karyotype/FISH/mtDNA/repeat-expansion testing: **not indicated** (point-mutation disorder).
- **Clinical criteria / classification:** meets **SCID / Omenn syndrome** diagnostic criteria (profound T-cell lymphopenia $\pm$ erythroderma, eosinophilia, high IgE, oligoclonal T cells); classified under IUIS "combined immunodeficiencies with associated/syndromic features."
- **Differential diagnosis:** other SCID/Omenn genes—**RAG1/RAG2, DCLRE1C (Artemis), LIG4, NHEJ1, DNA-PKcs/PRKDC, IL2RG, JAK3, IL7R, ADA, RMRP (cartilage-hair hypoplasia)**; and the related **POLD1- and POLD2-deficiency** syndromes (replication-stress CID;

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Distinguishing features: POLD3/POLD1/POLD2 disease uniquely couples SCID with

replicative stress + neurodevelopmental delay + sensorineural hearing loss + ectodermal signs. Also distinguish **POLD1 gain-of-function/MDPL** (progeroid, mandibular hypoplasia, lipodystrophy) and **POLD1/POLD2 proofreading-associated polyposis/cancer**, which are mechanistically different and **not** part of IMD122.

- **Screening:** detectable by **newborn SCID screening (TREC assay)** given profound T-cell lymphopenia; **carrier/prenatal/PGT** testing feasible once familial variant known.

11. Outcome / Prognosis

- **Survival/mortality: poor.** SCID is fatal in infancy without HSCT; even with HSCT at 6 months, one patient **died at age 4 years** from progressive neurological regression (P 38099988). Disease-specific mortality is high.
- **Morbidity/disability:** severe—recurrent/opportunistic infections, failure to thrive, **developmental disability, sensorineural deafness**, bronchiectasis (chronic lung damage).
- **Recovery potential:** HSCT can reconstitute immunity (hematopoietic compartment) but **does not correct the cell-intrinsic replication defect in CNS/cochlea/ectoderm**, so neurodevelopmental and auditory outcomes remain poor.
- **Prognostic factors:** age at HSCT, pre-transplant infection/organ damage, and—unique to this disorder—**severity of the non-hematopoietic replication defect** (neuro/auditory), which transplantation cannot rescue.
- **Prognostic biomarkers:** degree of replication stress / genomic instability (γ H2AX, S-phase defect) is a mechanistic correlate of severity (inferred).

12. Treatment

(No disease-specific approved therapy exists; management follows SCID/Omenn principles.)

- **Definitive: Allogeneic hematopoietic stem cell transplantation (HSCT)** — NCIT:C15431 (Hematopoietic Stem Cell Transplantation) / NCIT:C105867 (Allogeneic HSCT). Reconstitutes the immune system; performed at 6 months in the reported Omenn patient

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Does **not** address non-immune manifestations.

- **Supportive / prophylactic (standard SCID care, NCIT terms):**
- **Immunoglobulin replacement therapy (IVIG/SCIG)** — NCIT:C603 (Immunoglobulin Therapy).
- **Antimicrobial prophylaxis** — anti-*Pneumocystis* (co-trimoxazole), antifungal, antiviral prophylaxis.
- **Irradiated, CMV-safe, leukodepleted blood products**; avoidance of **live vaccines**.
- Nutritional support for feeding difficulties; management of erythroderma; **immunosuppression** (e.g., steroids/cyclosporine) to control Omenn immune dysregulation pre-HSCT.
- Rehabilitation: **hearing aids/cochlear implantation**, early developmental/physical/speech therapy, audiology follow-up.
- **Gene therapy / RNA / targeted / immunotherapy**: none available or in trials for POLD3 deficiency. In vitro, **WT POLD3 restoration rescues** the cellular phenotype

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providing proof-of-concept that gene correction could address the cell-intrinsic defect—**experimental/hypothetical only**.

- **Pharmacogenomics**: none specific; standard transplant-conditioning pharmacogenetics (e.g., busulfan, thiopurines) apply generically. Caution with **genotoxic conditioning/chemotherapy** given underlying DNA-repair/replication fragility (inferred; analogous to other DNA-repair SCIDs such as Artemis/LIG4, where reduced-toxicity conditioning is preferred).
- **Clinical trials (NCT)**: none identified specific to POLD3/IMD122.

13. Prevention

- **Primary prevention**: not preventable at the individual level (monogenic, congenital). **Genetic counseling** for consanguineous couples and families with an affected child (25% recurrence risk per pregnancy).

- **Secondary prevention / early detection: newborn SCID screening (TREC)** enables presymptomatic detection and **early pre-infection HSCT**, the key opportunity to improve immune outcomes. **Cascade testing** of at-risk relatives.
- **Genetic/reproductive prevention: carrier screening, prenatal diagnosis, and preimplantation genetic testing (PGT-M)** once the familial variant is known.
- **Tertiary prevention:** infection prophylaxis, avoidance of live vaccines and non-irradiated blood products, early audiologic and developmental intervention, pulmonary care to limit bronchiectasis progression.
- **Immunization/public-health/environmental measures:** not applicable beyond standard infection-control for immunocompromised infants.

14. Other Species / Natural Disease

- **Taxonomy / orthologs:** *POLD3* is conserved across eukaryotes. Mouse ***Pold3*** (NCBI Gene 69745; *Mus musculus*, NCBI:txid10090); orthologs in rat, zebrafish; the yeast functional homolog is **POL32** (*Saccharomyces cerevisiae*). *POLD3*/*POL32* is part of Pol δ and Pol ζ across species.
- **Natural disease in other species:** no naturally occurring *POLD3* disease reported in companion animals or wildlife (OMIA: none identified). Not a zoonosis; not transmissible.
- **Comparative biology:** the replication/genome-maintenance function of *POLD3*/*POL32* is **deeply evolutionarily conserved**, making model systems highly informative. Complete loss is lethal from yeast to mouse, underscoring essentiality.
- **Veterinary relevance:** none established.

15. Model Organisms

- **Mouse (*Mus musculus*, *Pold3*):**
- **Complete knockout (*Pold3*^{-/-}) is embryonic lethal at E6.5**

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"complete loss of *Pold3* (*Pold3*^{-/-}) resulted in early embryonic lethality at E6.5."

- **Inducible KO / blastocyst outgrowth / ESCs:** rapid DNA-damage response, massive apoptosis, **telomere loss, chromosome breaks, extended S phase, replicative stress, micronucleation, aneuploidy**; Pold3 acts via **53BP1, RIF1, ATR, ATM**.
- **Pold3^{+/-} (haploinsufficient)** mice develop **age-dependent** impaired DSB repair, telomere shortening and chromosome breaks (spermatocytes).
- **Phenotype recapitulation:** strongly models the **cell-intrinsic replication-stress/genomic-instability** mechanism of human disease; **limitation:** null is embryonic-lethal, so it cannot model the viable, organ-specific human syndrome—**hypomorphic knock-in alleles** (e.g., patient-equivalent) would be needed to recapitulate SCID + neuro/auditory features.
- **Cellular / in vitro models: patient-derived fibroblasts** (S-phase defect, γ H2AX DSB foci, rescued by WT POLD3 transduction —

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patient immune cells (TCR-repertoire/recombination assays). Yeast **pol32 Δ** is a classic genetic model of the accessory subunit.

- **Other organisms:** chicken DT40 and yeast systems have historically defined POLD3/POL32 roles in replication, translesion synthesis, and break-induced replication (background literature).
- **Resources:** MGI (Pold3), IMPC/IMSR, Alliance of Genome Resources, SGD (POL32), Cellosaurus (patient fibroblast lines).

Supported vs. Refuted Hypotheses

Supported: - IMD122 is caused by autosomal-recessive, biallelic hypomorphic *POLD3* missense variants

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38099988). - The mechanism is reduced Pol δ function → replicative stress / S-phase defect / DSB accumulation → impaired lymphopoiesis and TCR recombination → SCID/Omenn, with parallel injury to neural, cochlear, and ectodermal lineages

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29447390; 31449058). - POLD3 is essential and LoF-intolerant; only hypomorphic alleles are viable (gnomAD; mouse null lethality).

Refuted / excluded: - Not caused by environmental, infectious, or acquired/somatic factors; infections are a consequence. - Not a chromosomal/structural or repeat-expansion disorder. - IMD122 is **distinct** from POLD1 gain-of-function **MDPL** progeroid syndrome and from POLD1/POLD2 proofreading-associated polyposis/cancer.

Limitations and Future Directions

- **n=2** patients: frequencies, penetrance range, sex ratio, and full phenotypic spectrum are provisional; natural-history data are absent.
 - No omics (transcriptomic/proteomic/metabolomic) profiling, no registry, no clinical trials.
 - Unresolved: why certain tissues (cochlea, specific neural populations) are selectively vulnerable; whether reduced-toxicity HSCT conditioning improves outcomes; whether early gene-correction could rescue non-immune disease.
 - **Future:** patient-equivalent hypomorphic knock-in mouse models; iPSC-derived neural/cochlear organoids; identification of additional patients via GeneMatcher to define the phenotypic spectrum and genotype–phenotype correlations.
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Key References

- **P** 37030525
— Mehawej et al., 2023. *POLD3 deficiency is associated with severe combined immunodeficiency, neurodevelopmental delay, and hearing impairment*. (First case; p.Ile10Thr.)
- **P** 38099988
— Riestra et al., 2023. *Human Autosomal Recessive DNA Polymerase Delta 3 Deficiency Presenting as Omenn Syndrome*. (Second case; p.K373T; functional S-phase/DSB rescue.)
- **P** 31449058
— Conde et al., 2019 (J Clin Invest). *Polymerase δ deficiency causes syndromic immunodeficiency with replicative stress*. (POLD1/POLD2 parent syndrome.)

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— Zhou et al., 2018. *Pold3 is required for genomic stability and telomere integrity in embryonic stem cells and meiosis.* (Mouse model.)

- Database sources: OMIM 620869; MONDO:0971151 (Monarch); UniProt Q15054; gnomAD (POLD3 constraint); HPO annotations (OMIM 620869).

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